

STI's Solution for High Quantity Production of Stirling Coolers

Amr O'Baid, Andreas Fiedler, Abhijit Karandikar

Superconductor Technologies Incorporated
Santa Barbara, CA 93111

ABSTRACT

The mission of Superconductor Technologies Inc. (STI) is improving the quality of wireless. Through the use of its patented High Temperature Superconductor (HTS) technology, STI has created a cryogenic receiver front-end (CRFE) used by major wireless operators to enhance their network performance. The "Sapphire" cryocooler is a key component in the STI SuperLink product family. Since 1999, more than 4,000 Sapphire cryocoolers have been deployed with over 44 million hours of accumulated run time. The current generation of cryocooler has an efficiency of 16.7 W/W and a 6 W lift at 77 K with 100 W input power and 35°C heat rejection temperature.

STI's Sapphire cryocooler is not only suitable for wireless industry products, but for other applications where low cost, high efficiency, and field-proven reliability are decisive components for enabling product commercialization. The wireless communication industry demands products that are cost effective and extremely reliable, typically with MTBF requirements in the hundreds of thousands of hours. Field data confirms that the Sapphire cryocooler has met and even exceeded these requirements.

Performance data are presented along with reliability data from the field and in-house testing. Various methods of thermal management of the cryocooler in systems are also discussed.

INTRODUCTION

For seven years, STI has been producing Stirling cycle cryocoolers as part of their CRFE products for the wireless communication industry. The free piston, gas-bearing cryocooler is a critical component that contributes to the overall receiver system's lifetime and reliability. Cryocooler reliability data, therefore, is continuously tracked and updated. To date, the product has demonstrated an MTBF of greater than 500,000 hours based on reliability data from the field and in-house lifetest results. This has far surpassed STI's original goal of 50,000 hours. The extraordinary MTBF of > 500,000 hours has remained consistent over the last two years even with an increased number of deployed systems. The "Invisible Cooler," which is small, lightweight, and highly reliable has become a reality. STI's unique cryocooler design has a new name: "Sapphire."

Figure 1 shows STI's flagship product, the SuperLink Rx 850, one of over 4,000 maintenance-free SuperFilter and SuperLink Rx units deployed worldwide. In this figure, the Sapphire cryocooler is surrounded with a shroud to optimize forced-convection cooling of the unit using a fan.

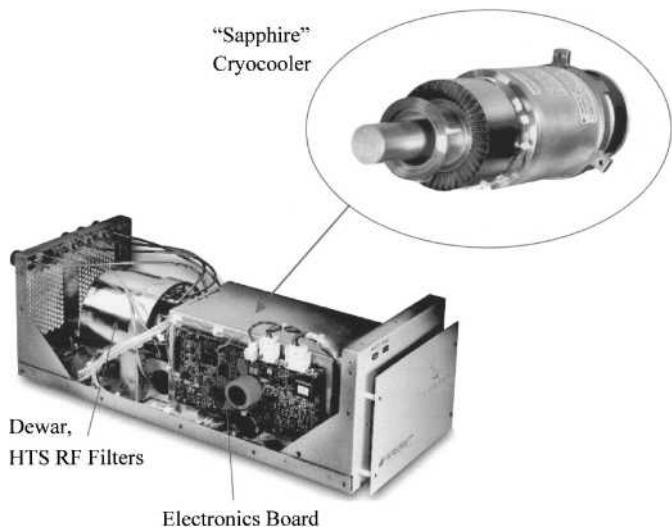


Figure 1. STI's flagship product, SuperLink Rx 850.

Cryocooler Performance

The Sapphire cryocooler is used in conjunction with a dewar and HTS RF filters based on “thin-film” technology that operates at a temperature of 77 K. The cooler has a heat load of 3.5 W from the dewar and is designed to operate over the complete ambient temperature range of -40°C to +60°C. The coldfinger temperature is regulated with the electronics board shown in Fig. 1.

STI's cryocooler has more than sufficient heat lift even under the worst-case conditions. The performance and physical characteristics of the Sapphire cryocooler are listed in Table 1. Figure 2 shows the heat lift of the Sapphire cryocooler as a function of input power for three different heat rejection temperatures.

RELIABILITY DATA FROM THE FIELD

STI has more than 4,000 units operating in the field. The combined run time is over 44 million hours. Of these units, 460 coolers have each > 26,280 accumulated hours (three years). Table 2 shows summary statistics for the STI units in the field. Figure 3 shows the accumulated run time and the number of coolers in the field as a function of time.

As part of STI's rigorous reliability and continuous improvement program, a thorough Pareto analysis is conducted on a regular basis to identify potential cryocooler design and reliability improvements. This resulted in cryocooler design changes in Q1 2002. Since that time over 2,600 coolers have been built, accumulating over 15 million hours in the field. Only three of these units have failed, and these failures were not wear-related.

Table 1. Performance and physical characteristics of the Sapphire cryocooler.

Heat lift	6 W of lift at 77K and 35°C heat reject temperature
Nominal input power	100 W input power at 6 W, 77 K and 35°C
Maximum input power	140 W, 60 Hz
Weight	7.5 lbs.
Dimensions	3.5 inches OD x 11.5 inches length
Operating orientation	Any
Ambient temperature	-40°C to +60°C
Vibration output	2.2 N with passive balancer